

A Related Work

LLM-based Human Simulation LLM agents have been increasingly adopted as effective proxies for humans in research fields such as sociology and economics (Xu et al., 2024; Horton, 2023; Gao et al., 2023b). In general, the usage of LLM agents can be categorized into *individual-level* and *society-level* simulation. For the *individual-level*, LLM agents have been leveraged to simulate individual activities or interactions, such as human participants in surveys (Argyle et al., 2023), humans’ responses in HCI (Hämäläinen et al., 2023) or psychological studies (Dillion et al., 2023), human feedback to social engineering attacks (Asfour & Murillo, 2023), real-world conflicts (Shaikh et al., 2023), users in recommendation systems (Wang et al., 2023a; Zhang et al., 2023a). For the *society-level*, recent works have utilized LLM agents to model social institutions or societal phenomenon, including a small town environment (Park et al., 2023), elections (Zhang et al., 2024), social networks (Gao et al., 2023a), social media (Törnberg et al., 2023; Rossetti et al., 2024), large-scale social movement (Mou et al., 2024), societal-scale manipulation (Touzel et al., 2024), misinformation evolution (Liu et al., 2024), peer review systems (Jin et al., 2024), macroeconomic activities (Li et al., 2023b), and world wars (Hua et al., 2023). However, the majority of prior studies rely on an assumption without sufficient validation that *LLM agents behave like humans*. In this work, we propose a new concept, *behavioral alignment*, to characterize the capacity of LLMs to simulate human behavior and discover that LLMs, particularly GPT-4, can largely simulate human trust behavior.

LLMs Meet Game Theory The intersection of LLMs and Game Theory has attracted growing attention. The motivation is generally two-fold. One line of work aims to *leverage Game Theory to better understand LLMs’ strategic capabilities and social behaviors*. For example, Akata et al. (2023); Fan et al. (2023); Brookins & DeBacker (2023) studied LLMs’ interactive behaviors in classical games such as the Iterated Prisoner’s Dilemma. Wang et al. (2023b); Lan et al. (2023); Light et al. (2023); Shi et al. (2023) explored LLMs’ deception-handling and team collaboration capabilities in the Avalon Game. Xu et al. (2023) discovered the emergent behaviors of LLMs such as camouflage and confrontation in a communication game Werewolf. Guo et al. (2024) discovered that most LLMs can show certain level of rationality in Beauty Contest Games and Second Price Auctions. Mukobi et al. (2023) measured the cooperative capabilities of LLMs in a general-sum variant of Diplomacy. Guo et al. (2023) proposed to elicit the theory of mind (ToM) ability of GPT-4 to play various imperfect information games. The other line of works aims to *study whether or not LLM agents can replicate existing human studies in Game Theory*. This direction is still in the initial stage and needs more efforts. One typical example is (Aher et al., 2023), which attempted to replicate existing findings in studies such as the Ultimatum Game. Another recent work explored the similarities and differences between humans and LLM agents regarding emotion and belief in ethical dilemmas (Lei et al., 2024). Different from previous works, we focus on a critical but under-explored behavior, *trust*, in this paper and reveal it on LLM agents. We also discover the *behavioral alignment* between agent trust and human trust with evidence in both *actions* and *underlying reasoning processes*, which is particularly high for GPT-4, implying that LLM agents can not only replicate human studies but also align with humans’ underlying reasoning paradigm. Our discoveries illustrate the great potential to simulate human trust behavior with LLM agents.

B Impact Statement

Our discoveries provide strong empirical evidence for validating the potential to simulate the trust behavior of humans with LLM agents, and pave the way for simulating more complex human interactions and social institutions where trust is an essential component.

Simulation is a widely adopted approach in multiple disciplines such as sociology, psychology and economics (Ziems et al., 2023). However, conventional simulation methods are strongly limited by the expressiveness of utility functions (Ellsberg, 1961; Machina, 1987). Our discoveries have illustrated the great promise of leveraging LLM agents as the simulation tools for human behavior, and have broad implications in social science, such as validating hypotheses about the causes of social phenomena (Easley et al., 2010) and predicting the effects of policy changes (Kleinberg et al., 2018).

Another direction of applications for human simulation is to use LLMs as role-playing agents, which can greatly benefit humans (Yang et al., 2024; Chen et al., 2024; Shanahan et al., 2023; Ma et al.,

2024). For example, Shaikh et al. (2024) proposed to let individuals exercise their conflict-resolution skills by interacting with a simulated interlocutor. Yue et al. (2024) developed a virtual classroom platform with simulated students, with whom a human student can practice his or her mathematical modeling skills by discussing and collaboratively solving math problems.

However, this paper also shows that some LLMs, especially the ones with a relatively small scale of parameters, are still deficient in accurately simulating human trust behavior, suggesting the potential to largely improve their behavioral alignment with humans. In addition, our paper also demonstrates the biases of LLM agents’ trust behavior towards specific genders and races, which sheds light on the potential risks in human behavior simulation and calls for more future research to mitigate them.

C Limitations and Future Works

In this paper, we leveraged an established framework in behavioral economics, Trust Games, to study the trust behavior of LLM agents, which simplifies real-world scenarios. More studies on LLM agents’ trust behavior in complex and dynamic environments are desired in the future. Also, trust behavior embraces both the actions and underlying reasoning processes. Thus, collective efforts from different backgrounds and disciplines such as behavioral science, cognitive science, psychology, and sociology are needed to gain a deeper understanding of LLM agents’ trust behavior and its relationship with human trust behavior.

D Additional Illustration for Experiments on Risk Perception

In the original human studies (Bohnet & Zeckhauser, 2004), participants are asked to directly indicate their Minimum Acceptable Probabilities (MAP) of trusting the trustee as P^* . Then, we can calculate Trust Rates (%) of the whole group of participants under different probability p . Specifically, when the probability p is higher than one participant’s P^* , we regard his or her decision as trusting the trustee. When the probability p is lower than one participant’s P^* , we regard his or her decision as not trusting the trustee. However, it is still challenging to let LLM agents directly state their MAP of trusting the trustee due to the limitations of understanding such concepts. Then, we conducted 10 groups of experiments with p from 0.1 to 1.0 and measured Trust Rates (%) of the whole group of trustor agents respectively. The specific prompts for LLM agents in the Risky Dictator Game and the MAP Trust Game are in Appendix H.2.

E Statistical Testing

LLM	<i>p</i> -value
text-davinci-003	0.03
Llama-2-13b	0.03
Vicuna-13b-v1.3	0.35
Vicuna-7b-v1.3	0.50
GPT-3.5-turbo-0613	0.42
Vicuna-33b-v1.3	0.33
Llama-2-70b	0.03
GPT-3.5-turbo-instruct	0.10
GPT-4	0.05

Table 1: **Statistical Testing of The Change of Amount Sent for LLM Agents between the Trust Game and the Dictator Game (Figure 3).** “*p*-value” indicates the statistical significance of the change and is calculated with an One-Tailed Independent Samples t-test.